Supplemental Materials

for

Noyes, et al.

Supplemental Tables appear in the accompanying spreadsheet file.

Supplemental Tables

Table S1: Collection sites.

Municipality	Latitude	Longitude	Host plant	Sample size in January	Sample size in July	Proportion of green space	Urbanicity index
Lake Wales FL	27.9594	-81.5941	*K. paniculata*	2	3	0.7579	-3.385
Lake Wales FL	27.8438	-81.5853	*K. paniculata*	0	25	0.4180	-2.621
Islamorada FL	24.9501	-80.5957	*C. corindum*	0	5	0.0539	-2.517
Bartow FL	27.9549	-81.7708	*K. paniculata*	5	1	0.2041	-1.988
Lakeland FL	28.0927	-81.9536	*K. paniculata*	0	4	0.1697	-0.014
Lake Alfred FL	28.0918	-81.7231	*K. paniculata*	6	14	0.4207	1.107
Winter Haven FL	28.0433	-81.7615	*K. paniculata*	3	0	0.2931	1.312
Lake Bentley FL	28.0211	-81.9322	*K. paniculata*	3	0	0.5783	1.403
Fort Lauderdale FL	26.1884	-80.1720	*K. paniculata*	10	9	0.0806	1.697
Lakeland FL	28.0334	-81.9552	*K. paniculata*	6	10	0.2920	1.758
Lehigh Acres FL	26.6173	-81.6897	*K. paniculata*	5	0	0.4459	1.763
Delray Beach FL	26.4749	-80.0912	*K. paniculata*	6	19	0.0929	1.916
Kendall Lakes FL	25.7139	-80.4203	*K. paniculata*	8	0	0.2868	2.102

Table S2: DADA2 Denoise samples statistics. The number of input reads, those passing quality filter and their percentage, the number denoised and merged and their percentage, the number of non-chimeric sequences and their percentage.

This table is very large. Please refer to the accompanying spreadsheet.

Table S3: Core microbiome in each host plant cohort.

Host plant	ASV ID	Mean abundance	Prevalence	Phylum	Class	Order	Family	Genus
C. corindum	02502141f0a1bf07cc6d6c60b94433f5	0.278790	1	Proteobacteria	Gammaproteobacteria	Enterobacterales	Enterobacteriaceae	NA
C. corindum	5f9403f34de2d6658df09b913aa13184	0.242246	1	Proteobacteria	Alphaproteobacteria	NA	NA	NA
C. corindum	7364a9d6bfb4a993fabfa2e22252a18d	0.230191	1	Proteobacteria	Gammaproteobacteria	Enterobacterales	Enterobacteriaceae	NA
C. corindum	8084aa68823c09679d87c31776e22478	0.150427	1	Firmicutes	Bacilli	Lactobacillales	Streptococcaceae	Lactococcus
K. paniculata	9f8c37a25eca1c8caec4c05d14cd7cdf	0.661537	0.992	Proteobacteria	Alphaproteobacteria	Rhizobiales	Rhizobiaceae	NA
K. paniculata	8084aa68823c09679d87c31776e22478	0.135416	0.841	Firmicutes	Bacilli	Lactobacillales	Streptococcaceae	Lactococcus

Table S4: Sample alpha diversity metrics.

This table is very large. Please refer to the accompanying spreadsheet.

Table S5: Alpha test statistics. Binary categorical factors were tested using the Wilcoxon rank sum test, producing a W statistic. Continuous factors were modeled in linear mixed effects models with collection site as a random intercept, producing t statistics. Associated p-values are listed without correction.

metric	factor	W	p	
observed_ASVs	sex	2149.5	0.07717	
shannon	sex	2500.5	0.7176	
simpson	sex	2593.5	0.9952	
observed_ASVs	morph	2074.0	0.4926	
shannon	morph	2112.0	0.6029	
simpson	morph	2108.0	0.5909	
observed_ASVs	host_plant	316.0	0.06921	
shannon	host_plant	352.0	0.01435	*
simpson	host_plant	356.0	0.01177	*
observed_ASVs	month	1491.0	0.0005031	***
shannon	month	1306.5	0.00001957	***
simpson	month	1337.5	0.00003538	***
metric	**factor**	**t**	**p**	
observed_ASVs	urbanicity	-0.846	0.423	
shannon	urbanicity	-1.544	0.182308	
simpson	urbanicity	-1.553	0.174039	
observed_ASVs	green_space	0.534	0.60365	
shannon	green_space	0.534	0.60365	
simpson	green_space	-1.369	0.198404	

Table S6: PERMANOVA test statistics.

Distance matrix	factors	df	SS	R^2	F	p	
Bray-Curtis	sex	1	0.1469	0.00491	0.7919	0.5005	
	morph	2	0.2861	0.00956	0.7714	0.554446	
	host_plant	1	2.0693	0.06914	11.1585	0.000999	***
	month	1	0.3815	0.01275	2.0574	0.090909	.
	green_space	1	0.8789	0.02937	4.7394	0.001998	**
	Residual	137	25.4065	0.84892			
	Total	143	29.928	1			
unweighted UniFrac	sex	1	0.412	0.00937	1.4688	0.080919	.
	morph	2	0.741	0.01687	1.3215	0.105894	
	host_plant	1	1.819	0.04138	6.4842	0.000999	***
	month	1	1.542	0.03509	5.4979	0.000999	***
	green_space	1	0.994	0.0226	3.5418	0.000999	***
	Residual	137	38.434	0.87437			
	Total	143	43.956	1			
weighted UniFrac	sex	1	0.00017	0.00009	0.0137	0.94805	
	morph	2	0.01561	0.0087	0.6337	0.56543	
	host_plant	1	0.02669	0.01488	2.1675	0.1009	
	month	1	0.00852	0.00475	0.6918	0.46853	
	green_space	1	0.04483	0.025	3.6411	0.04296	*
	Residual	137	1.6868	0.94068			
	Total	143	1.79317	1			
Bray-Curtis	host_plant	1	2.6586	0.14451	14.865	0.000999	***
	Residual	88	15.7391	0.85549			
	Total	89	18.3977	1			
unweighted UniFrac	host_plant	1	1.7687	0.07362	6.9929	0.000999	***
	Residual	88	22.2572	0.92638			
	Total	89	24.0258	1			
weighted UniFrac	host_plant	1	0.03196	0.03447	3.1414	0.04595	*
	Residual	88	0.89516	0.96553			
	Total	89	0.92712	1			
Bray-Curtis	month	1	0.4096	0.01536	2.1375	0.1039	
	Residual	137	26.2553	0.98464			
	Total	138	26.6649	1			
unweighted UniFrac	month	1	1.679	0.04059	5.7962	0.000999	***
	Residual	137	39.683	0.95941			
	Total	138	41.361	1			
weighted UniFrac	month	1	0.0096	0.00548	0.7547	0.4236	
	Residual	137	1.742	0.99452			
	Total	138	1.7516	1			

Table S7: ANCOM-BC2 Results.

This table is very large. Please refer to the accompanying spreadsheet.

Table S8: ALDEx2 Results.

This table is very large. Please refer to the accompanying spreadsheet.

Table S9: MaAsLin3 Results.

This table is very large. Please refer to the accompanying spreadsheet.

Table S10: Consensus of differential abundance analysis methods.

This table is very large. Please refer to the accompanying spreadsheet.

Table S11: PERMDISP Results.

Distance matrix	factors	Df	SS	MSS	F	Pr(>F)	
Bray-Curtis	Groups	2	1.4526	0.72629	7.3334	0.00104	**
	Residuals	136	13.4691	0.09904			
unweighted UniFrac	Groups	2	0.4246	0.212289	7.086	0.00155	**
	Residuals	136	4.0744	0.029959			
weighted UniFrac	Groups	2	0.10743	0.053715	6.3726	0.00223	**
	Residuals	136	1.14635	0.008429			

Supplemental Figures

Figure S1

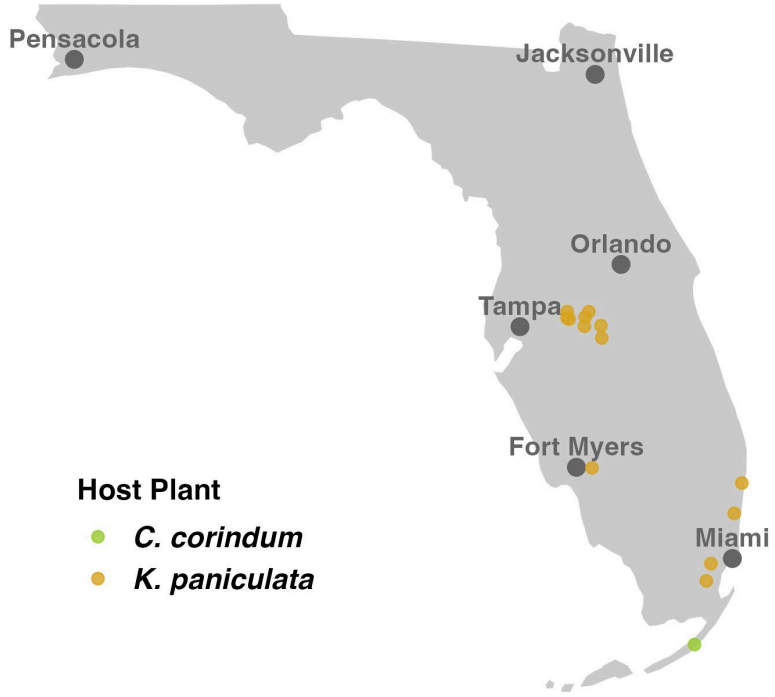


Figure S1: Map of collections sites in this study.

Figure S2

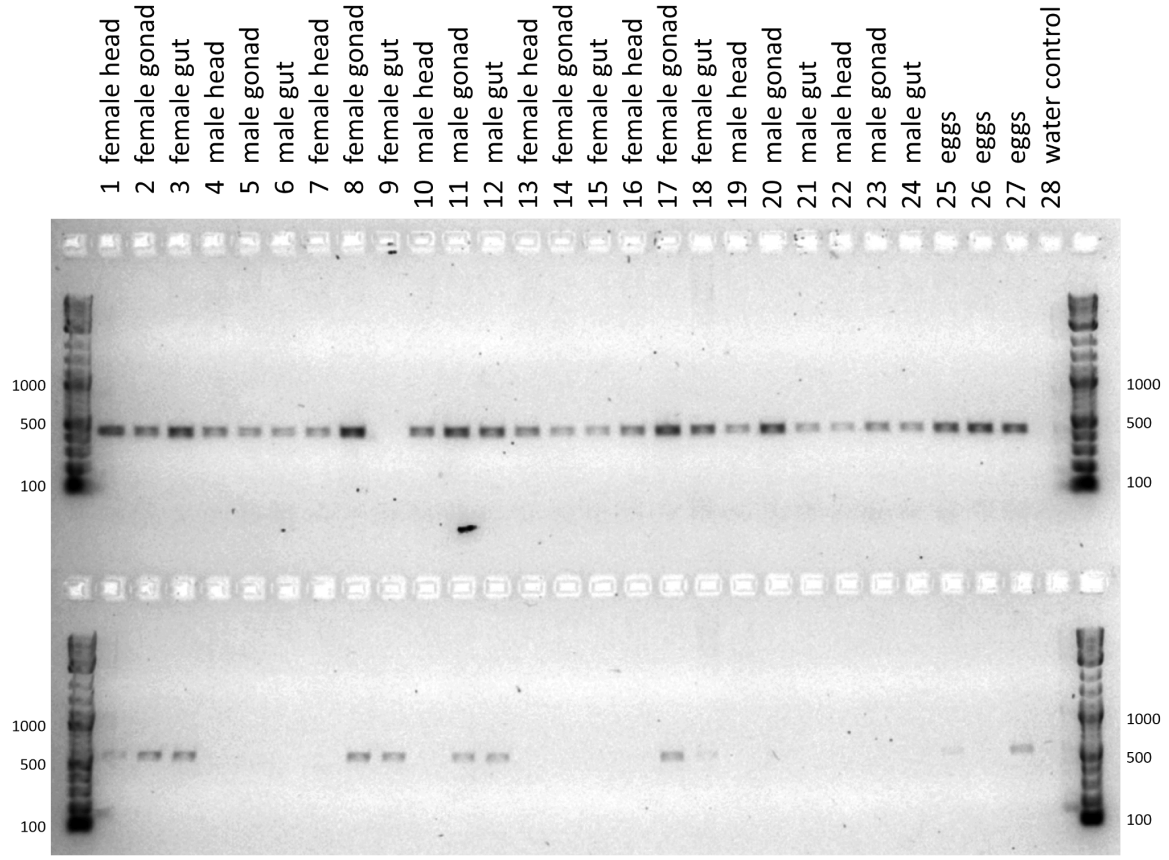


Figure S2: PCR screen for the presence of *Wolbachia* in *J. haematoloma* tissue. Template DNA from the head, gonads and gut of 4 female and 4 male adults, as well as 3 separate pools of eggs were used in PCRs targeting *Wolbachia*-specific sequences. Each template was tested with two primer pairs. PCR rxn conditions. Primers W-Spec-16S-F/R (Werren & Windsor 2000; top row) produce an expected product of 438 bp. All samples except one female gut sample produced the expected product. Primers ftsZ-F/R (Zha et al. 2014; bottom row) produce an expected product of 570 bp. Only some individuals appeared to be positive for *Wolbachia* based on this primer pair. All sampled tissues showed evidence of *Wolbachia* in at least one individual. Negative (water) controls were consistently negative for amplification with both primers pairs. Product sizes were determined by comparison to be New England BioLabs 1-kb Plus DNA ladder (outside lanes).

Figure S3

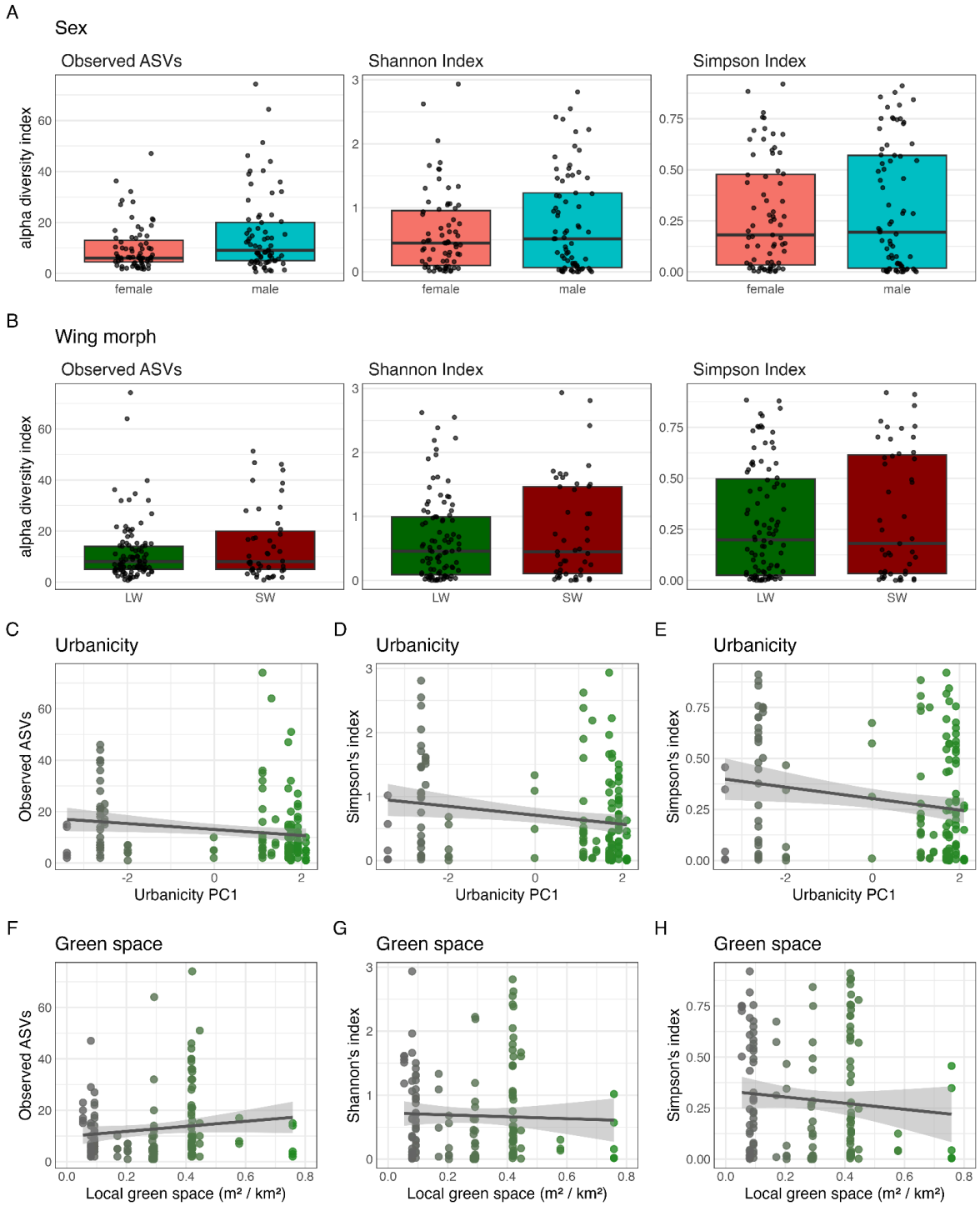


Figure S3: Non-significant factors in *J. haematoloma* gut microbiota alpha diversity. (A-B) Box plots displaying within-sample taxonomic richness and diversity. Each panel plots observed ASVs, Shannon index and Simpson's index. Center lines represent the median, box limits

indicate the upper and lower quartiles, and whiskers extend to 1.5 times the interquartile range. Statistical significance was tested using Wilcoxon rank-sum tests (Table S5). Factors with significant association to alpha diversity are shown in Figure 3. **(A)** Alpha diversity comparisons by sex. Females: n = 71; males = 73. **(B)** Alpha diversity comparisons between wing morphs: long-winged (LW, n = 95) and short-winged (SW, n = 47) adults. **(C-H)** Continuous factors are shown in scatterplots with a best-fit line (black) bounded by a 95% confidence interval (gray fill, n = 139). **(C-E)** Urbanicity, calculated as the first principle component from an analysis of land cover data at each collection site. **(F-H)** Proportion of local green space.

Figure S4

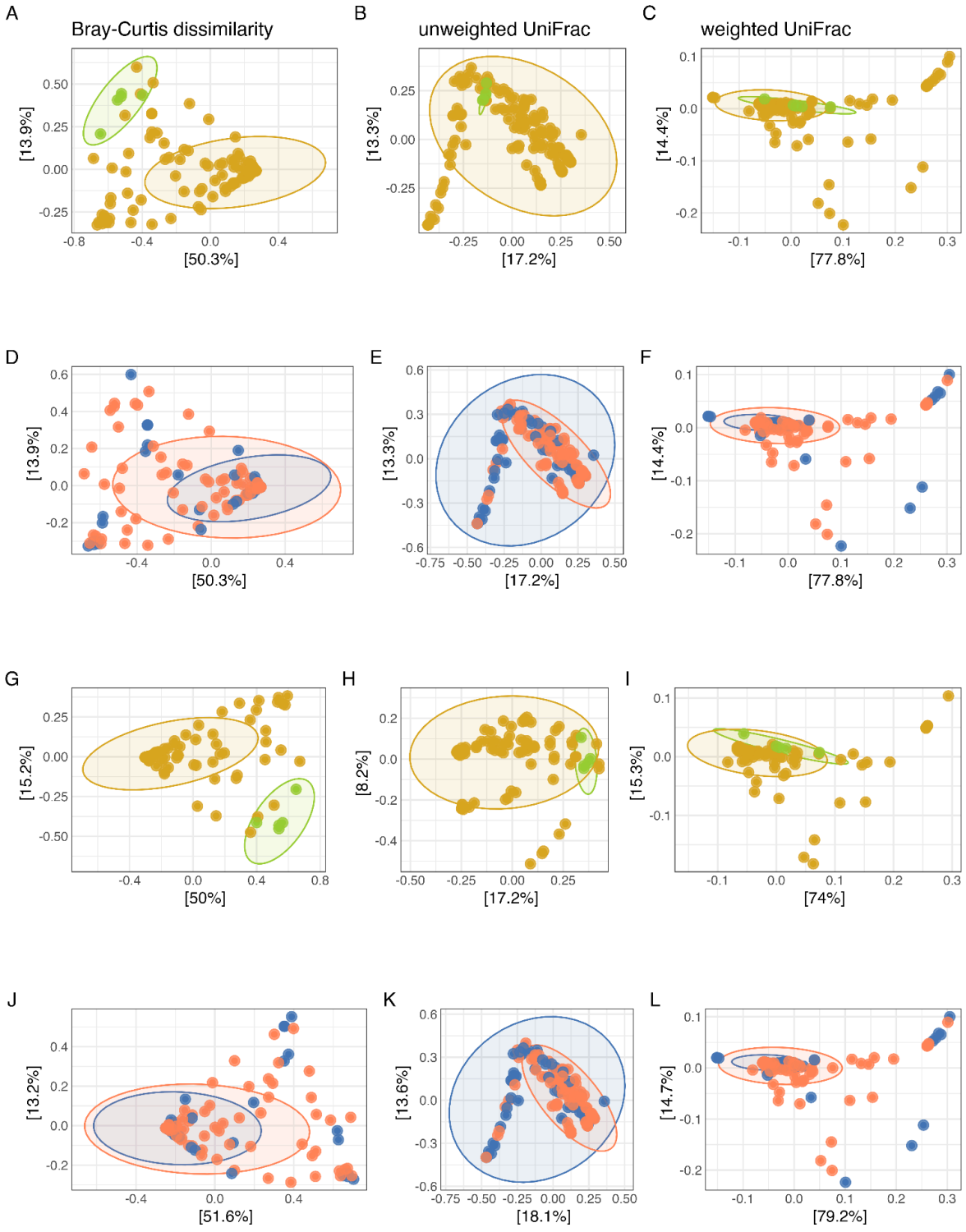


Figure S4: Principle coordinate analysis (PCoA) plots. Panels display the first and second PCoA axes for Bray-Curtis (A, D, G, J), unweighted UniFrac (B, E, H, K), and weighted

UniFrac (C, F, I, L) distances. Host plant association is highlighted in panels A-C and G-I with samples from the ancestral host plant *Cardiospermum corindum* in green and the introduced host plant *Koelreuteria paniculata* in orange. Samples collected in different seasons are color-coded in panels D-F and J-L, with January samples in dark blue and July samples in light red. PCoA was initially applied to the complete dataset (A-F). To address the unbalanced nature of sample collection, we also examined the effects of these factors in data subsets. Host plant effects were examined only among July samples (G-I) and seasonal effects were examined only among samples collected from *K. paniculata* (J-L). Shaded ellipses represent 95% confidence intervals assuming a multivariate distribution (`stat_ellipse`).